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🔗 Llama 2 is here - get it on Hugging Face

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🔗 Introduction

Llama 2 is a family of state-of-the-art open-access large language models released by Meta today, and we're excited to fully support the launch with comprehensive integration in Hugging Face. Llama 2 is being released with a very permissive community license and is available for commercial use. The code, pretrained models, and fine-tuned models are all being released today 🧨

We've collaborated with Meta to ensure smooth integration into the Hugging Face ecosystem. You can find the 12 open-access models (3 base models & 3 fine-tuned ones with the original Meta checkpoints, plus their corresponding transformers models) on the Hub. Among the features and integrations being released, we have:

- [Models on the Hub](#) with their model cards and license.
- [Transformers integration](#)
- Examples to fine-tune the small variants of the model with a single GPU
- Integration with [Text Generation Inference](#) for fast and efficient production-ready inference
- Integration with Inference Endpoints

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🔗 Why Llama 2?

The Llama 2 release introduces a family of pretrained and fine-tuned LLMs, ranging in scale from 7B to 70B parameters (7B, 13B, 70B). The pretrained models come with significant improvements over the Llama 1 models, including being trained on 40% more tokens, having a much longer context length (4k tokens 🤖), and using grouped-query attention for fast inference of the 70B model 🔥!

However, the most exciting part of this release is the fine-tuned models (Llama 2-Chat), which have been optimized for dialogue applications using [Reinforcement Learning from Human Feedback \(RLHF\)](#). Across a wide range of helpfulness and safety benchmarks, the Llama 2-Chat models perform better than most open models and achieve comparable performance to ChatGPT according to human evaluations. You can read the paper [here](#).

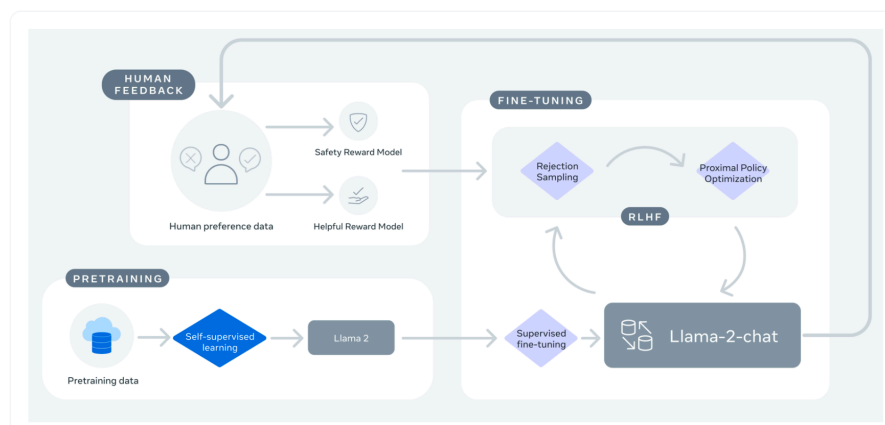


image from [Llama 2: Open Foundation and Fine-Tuned Chat Models](#)

If you've been waiting for an open alternative to closed-source chatbots, Llama 2-Chat is likely your best choice today!

Model	License	Commercial use?	Pretraining length [tokens]	Leaderboard score
Falcon-7B	Apache 2.0	✓	1,500B	44.17
MPT-7B	Apache 2.0	✓	1,000B	47.24
Llama-7B	Llama license	✗	1,000B	45.65

Model	License	Commercial use?	Pretraining length [tokens]	Leaderboard score
Llama-2-7B	Llama 2 license	✓	2,000B	50.97
Llama-33B	Llama license	✗	1,500B	-
Llama-2-13B	Llama 2 license	✓	2,000B	55.69
mpt-30B	Apache 2.0	✓	1,000B	52.77
Falcon-40B	Apache 2.0	✓	1,000B	58.07
Llama-65B	Llama license	✗	1,500B	61.19
Llama-2-70B	Llama 2 license	✓	2,000B	67.87
Llama-2-70B-chat	Llama 2 license	✓	2,000B	62.4

Note: the performance scores shown in the table below have been updated to account for the new methodology introduced in November 2023, which added new benchmarks. More details in [this post](#).

[Demo](#)

You can easily try the 13B Llama 2 Model in [this Space](#) or in the playground embedded below:



Loading...

huggingface-projects/llama-2-13b-chat built with Gradio.
Hosted on  Spaces

To learn more about how this demo works, read on below about how to run inference on Llama 2 models.

[Inference](#)

In this section, we'll go through different approaches to running inference of the Llama 2 models. Before using these models, make sure you have requested access to one of the models in the official [Meta Llama 2](#) repositories.

Note: Make sure to also fill the official Meta form. Users are provided access to the repository once both forms are filled after few hours.

🔗 Using transformers

With transformers [release 4.31](#), one can already use Llama 2 and leverage all the tools within the HF ecosystem, such as:

- training and inference scripts and examples
- safe file format (safetensors)
- integrations with tools such as bitsandbytes (4-bit quantization) and PEFT (parameter efficient fine-tuning)
- utilities and helpers to run generation with the model
- mechanisms to export the models to deploy

Make sure to be using the latest transformers release and be logged into your Hugging Face account.

```
pip install transformers
huggingface-cli login
```

In the following code snippet, we show how to run inference with transformers. It runs on the free tier of Colab, as long as you select a GPU runtime.

```
from transformers import AutoTokenizer
import transformers
import torch

model = "meta-llama/Llama-2-7b-chat-hf"

tokenizer = AutoTokenizer.from_pretrained(model)
pipeline = transformers.pipeline(
    "text-generation",
    model=model,
    torch_dtype=torch.float16,
    device_map="auto",
)

sequences = pipeline(
    'I liked "Breaking Bad" and "Band of Brothers". Do you have any re
    do_sample=True,
    top_k=10,
    num_return_sequences=1,
    eos_token_id=tokenizer.eos_token_id,
    max_length=200,
)
```

```
for seq in sequences:
    print(f"Result: {seq['generated_text']}")
```

Result: I liked "Breaking Bad" and "Band of Brothers". Do you have any Answer:

Of course! If you enjoyed "Breaking Bad" and "Band of Brothers," here

1. "The Sopranos" - This HBO series is a crime drama that explores the
2. "The Wire" - This HBO series is a gritty and realistic portrayal of
3. "Mad Men" - Set in the 1960s, this AMC series follows the lives of

And although the model has *only* 4k tokens of context, you can use techniques supported in transformers such as rotary position embedding scaling ([tweet](#)) to push it further!

🔗 Using text-generation-inference and Inference Endpoints

[Text Generation Inference](#) is a production-ready inference container developed by Hugging Face to enable easy deployment of large language models. It has features such as continuous batching, token streaming, tensor parallelism for fast inference on multiple GPUs, and production-ready logging and tracing.

You can try out Text Generation Inference on your own infrastructure, or you can use Hugging Face's [Inference Endpoints](#). To deploy a Llama 2 model, go to the [model page](#) and click on the [Deploy -> Inference Endpoints](#) widget.

- For 7B models, we advise you to select "GPU [medium] - 1x Nvidia A10G".
- For 13B models, we advise you to select "GPU [xlarge] - 1x Nvidia A100".
- For 70B models, we advise you to select "GPU [2xlarge] - 2x Nvidia A100" with bitsandbytes quantization enabled or "GPU [4xlarge] - 4x Nvidia A100"

Note: You might need to request a quota upgrade via email to api-enterprise@huggingface.co to access A100s

You can learn more on how to [Deploy LLMs with Hugging Face Inference Endpoints in our blog](#). The [blog](#) includes information about supported hyperparameters and how to stream your response using Python and Javascript.

🔗 Fine-tuning with PEFT

Training LLMs can be technically and computationally challenging. In this section, we look at the tools available in the Hugging Face ecosystem to efficiently train Llama 2 on simple hardware and show how to fine-tune the 7B version of Llama 2 on a single NVIDIA T4 (16GB - Google Colab). You can learn more about it in the [Making LLMs even more accessible blog](#).

We created a [script](#) to instruction-tune Llama 2 using QLoRA and the [SFTTrainer](#) from [trl](#).

An example command for fine-tuning Llama 2 7B on the [timdettmers/openassistant-guanaco](#) can be found below. The script can merge the LoRA weights into the model weights and save them as safetensor weights by providing the `merge_and_push` argument. This allows us to deploy our fine-tuned model after training using text-generation-inference and inference endpoints.

First pip install `trl` and clone the script:

```
pip install trl
git clone https://github.com/lvwerra/trl
```

Then you can run the script:

```
python trl/examples/scripts/sft_trainer.py \
  --model_name meta-llama/Llama-2-7b-hf \
  --dataset_name timdettmers/openassistant-guanaco \
  --load_in_4bit \
  --use_peft \
  --batch_size 4 \
  --gradient_accumulation_steps 2
```

🔗 How to Prompt Llama 2

One of the unsung advantages of open-access models is that you have full control over the system prompt in chat applications. This is essential to specify the behavior of your chat assistant –and even imbue it with some personality–, but it's unreachable in models served behind APIs.

We're adding this section just a few days after the initial release of Llama 2, as we've had many questions from the community about how to prompt the models and how to change the system prompt. We hope this helps!

The prompt template for the first turn looks like this:

```
<s>[INST] <<SYS>>
{{ system_prompt }}
<</SYS>>

{{ user_message }} [/INST]
```

This template follows the model's training procedure, as described in [the Llama 2 paper](#). We can use any `system_prompt` we want, but it's crucial that the format matches the one used during training.

To spell it out in full clarity, this is what is actually sent to the language model when the user enters some text (There's a llama in my garden 🐼 What should I do?) in [our 13B chat demo](#) to initiate a chat:

```
<s>[INST] <<SYS>>
You are a helpful, respectful and honest assistant. Always answer as h

If a question does not make any sense, or is not factually coherent, e
<</SYS>>

There's a llama in my garden 🐼 What should I do? [/INST]
```

As you can see, the instructions between the special <<SYS>> tokens provide context for the model so it knows how we expect it to respond. This works because exactly the same format was used during training with a wide variety of system prompts intended for different tasks.

As the conversation progresses, *all* the interactions between the human and the "bot" are appended to the previous prompt, enclosed between [INST] delimiters. The template used during multi-turn conversations follows this structure (🐼 h/t [Arthur Zucker](#) for some final clarifications):

```
<s>[INST] <<SYS>>
{{ system_prompt }}
<</SYS>>

{{ user_msg_1 }} [/INST] {{ model_answer_1 }} </s><s>[INST] {{ user_ms
```

The model is stateless and does not "remember" previous fragments of the conversation, we must always supply it with all the context so the conversation can continue. This is the reason why **context length** is a very important parameter to maximize, as it allows for longer conversations and larger amounts of information to be used.

🔗 Ignore previous instructions

In API-based models, people resort to tricks in an attempt to override the system prompt and change the default model behaviour. As imaginative as these solutions are, this is not necessary in open-access models: anyone can use a different prompt, as long as it follows the format described above. We believe that this will be an important tool for researchers to study the impact of prompts on both desired and unwanted characteristics. For example, when people [are surprised with absurdly cautious generations](#), you can explore whether maybe [a different prompt would work](#). (🐼 h/t [Clémentine Fourrier](#) for the links to this example).

In our [13B](#) and [7B](#) demos, you can easily explore this feature by disclosing the "Advanced Options" UI and simply writing your desired instructions. You can also

duplicate those demos and use them privately for fun or research!

🔗 Additional Resources

- [Paper Page](#)
- [Models on the Hub](#)
- [Leaderboard](#)
- [Meta Examples and recipes for Llama model](#)
- [Chat demo \(7B\)](#)
- [Chat demo \(13B\)](#)
- [Chat demo \(70B\) on TGI](#)

🔗 Conclusion

We're very excited about Llama 2 being out! In the incoming days, be ready to learn more about ways to run your own fine-tuning, execute the smallest models on-device, and many other exciting updates we're preparing for you!

More Articles from our Blog



Llama 3.1

Multilingual 405B, 70B & 8B and 128k context

Llama 3.1 - 405B, 70B & 8B with multilinguality and long context

By philschmid July 23, 2024 • 🔍 108



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